

Project 3: SQL Data Analysis Report

DecodeLabs Data Analytics Internship

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Tools Used: MySQL | SQL | GitHub

1. Project Overview

The objective of this project was to use SQL queries to extract meaningful insights from a sales dataset. By applying SQL techniques such as filtering, grouping, sorting, and aggregation, the analysis aimed to answer key business questions and support data-driven decision-making.

The project focused on transforming raw transactional data into actionable business intelligence through structured SQL queries.

2. Methodology

The following SQL concepts were applied throughout the analysis:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- COUNT()
- SUM()
- AVG()

The analysis focused on:

- Revenue performance
- Product performance
- Payment method preferences
- Referral source effectiveness
- Order status distribution
- Customer purchasing behavior

3. Analysis and Findings

Total Revenue Analysis

SQL queries were used to calculate the total revenue generated from all transactions and determine the average order value.

Finding

The dataset generated substantial revenue, with customer spending varying across different products and payment methods.

	total_revenue
▶	1264761.96

Total revenue generated across all transactions

	average_order_value
▶	1053.97

Average order value

Product Performance Analysis

Products were grouped and analyzed based on quantity sold and total revenue generated.

Finding

- Chairs recorded the highest sales volume.
- Chairs also generated the highest revenue.
- Phones generated the lowest revenue among all products.

This indicates that Chairs were the strongest-performing product category in the dataset.

	Product	quantity_sold
▶	Chair	562
	Printer	542
	Laptop	535
	Desk	508
	Tablet	497
	Monitor	480
	Phone	411

Quantity sold by product

	Product	revenue
▶	Chair	195620.11
	Printer	195612.61
	Laptop	192126.56
	Tablet	186568.95
	Monitor	175651.41
	Desk	167459.93
	Phone	151722.39

Payment Method Analysis

Payment methods were analyzed to determine customer preferences and revenue contribution.

Finding

- Online payments recorded the highest number of transactions.
- Credit Card transactions generated the highest revenue.

This suggests that customers using Credit Cards tend to make higher-value purchases.

	PaymentMethod	total_orders
▶	Online	258
	Cash	246
	Credit Card	234
	Debit Card	232
	Gift Card	230

Total orders by payment method

	PaymentMethod	revenue
▶	Credit Card	263847.63
	Online	262442.94
	Cash	259786.29
	Gift Card	246323.92
	Debit Card	232361.18

Revenue by payment method

Referral Source Analysis

Referral channels were analyzed to determine their effectiveness in attracting customers and generating revenue.

Finding

- Instagram generated the highest number of orders.
- Instagram also generated the highest revenue.
- Referral generated the lowest revenue.

This suggests that Instagram was the most effective marketing channel within the dataset.

	ReferralSource	total_orders
▶	Instagram	259
	Email	250
	Google	241
	Facebook	228
	Referral	222

Total orders by referral source

	referralSource	revenue
▶	Instagram	275285.45
	Email	261808.55
	Google	250441.48
	Facebook	250410.9
	Referral	226815.58

Revenue by referral source

Order Status Analysis

Order statuses were analyzed to understand operational performance and revenue distribution.

Finding

- Cancelled orders recorded the highest order count.
- Cancelled orders also represented the highest revenue value among order statuses.

This highlights a potential area of revenue leakage that requires further investigation.

	OrderStatus	total_orders
▶	Cancelled	250
	Returned	247
	Pending	237
	Shipped	235
	Delivered	231

Total orders by order status

	OrderStatus	revenue
▶	Cancelled	276396.21
	Pending	256328.15
	Shipped	246159.58
	Returned	243277.7
	Delivered	242600.32

Revenue by order status

4. Key Insights

- **Insight 1:** Chairs were the best-performing product, generating both the highest sales volume and highest revenue.
- **Insight 2:** Instagram was the strongest acquisition channel, contributing the highest number of orders and revenue.
- **Insight 3:** Online payments were the most popular payment method, while Credit Card transactions generated the highest revenue.
- **Insight 4:** Order cancellations represented a significant revenue risk and may be negatively impacting business performance.
- **Insight 5:** Customer spending patterns varied across payment methods, products, and referral sources, highlighting opportunities for targeted business strategies.

5. Recommendations

Grow Instagram Investment

Instagram generated both the highest order volume and the highest revenue among all referral channels. The business should increase investment in this channel while continuing to leverage the strategies driving its success.

Investigate and Reduce Order Cancellations

Cancelled orders account for the highest order count and revenue value, representing a significant risk to overall revenue. A root-cause analysis should be conducted to identify the factors driving cancellations.

Prioritize Top-Performing Products

Chairs, Printers, and Laptops are the strongest contributors to sales and revenue and should be prioritized in inventory planning and promotional campaigns.

Encourage Credit Card Usage

Although Online payments are the most frequently used payment method, Credit Card transactions generate the highest revenue. Offering incentives for credit card usage may increase average order value.

Optimize Marketing Resources

Lower-performing referral channels should be reviewed, with consideration given to reallocating marketing resources toward higher-performing channels.

6. Conclusion

This SQL analysis successfully transformed transactional sales data into meaningful business insights. Through the use of SQL queries and aggregation techniques, key patterns in revenue generation, product performance, customer behavior, and marketing effectiveness were identified.

The findings provide valuable recommendations that can help improve marketing performance, reduce revenue loss, and support data-driven decision-making.